User Tip: *When placing multiple layers of roller-compacted concrete, lay the second layer on the first layer within one hour to create an adequate bond between the layers.*

User Tip: *When paving at a depth of 15 cm (6") or more of roller-compacted concrete, be sure to activate the screed counterbalance system. Taking some weight off the screed will help the screed float when paving thick layers. Also, be sure to add additional angle of attack to the main screed. Turn the thickness screws on both sides of the screed into the plus range to add angle of attack to the main screed. Adjust the counterbalance pressure and the angle of attack until the texture of the layer is correct at the desired thickness.*



Use the screed counterbalance system to help the screed float properly when laying down thick layers of roller-compacted concrete.

Roller-compacted concrete tends to move laterally when subjected to compaction forces. To avoid lateral displacement, try to avoid constructing unconfined edges. If that option is impractical, simply:

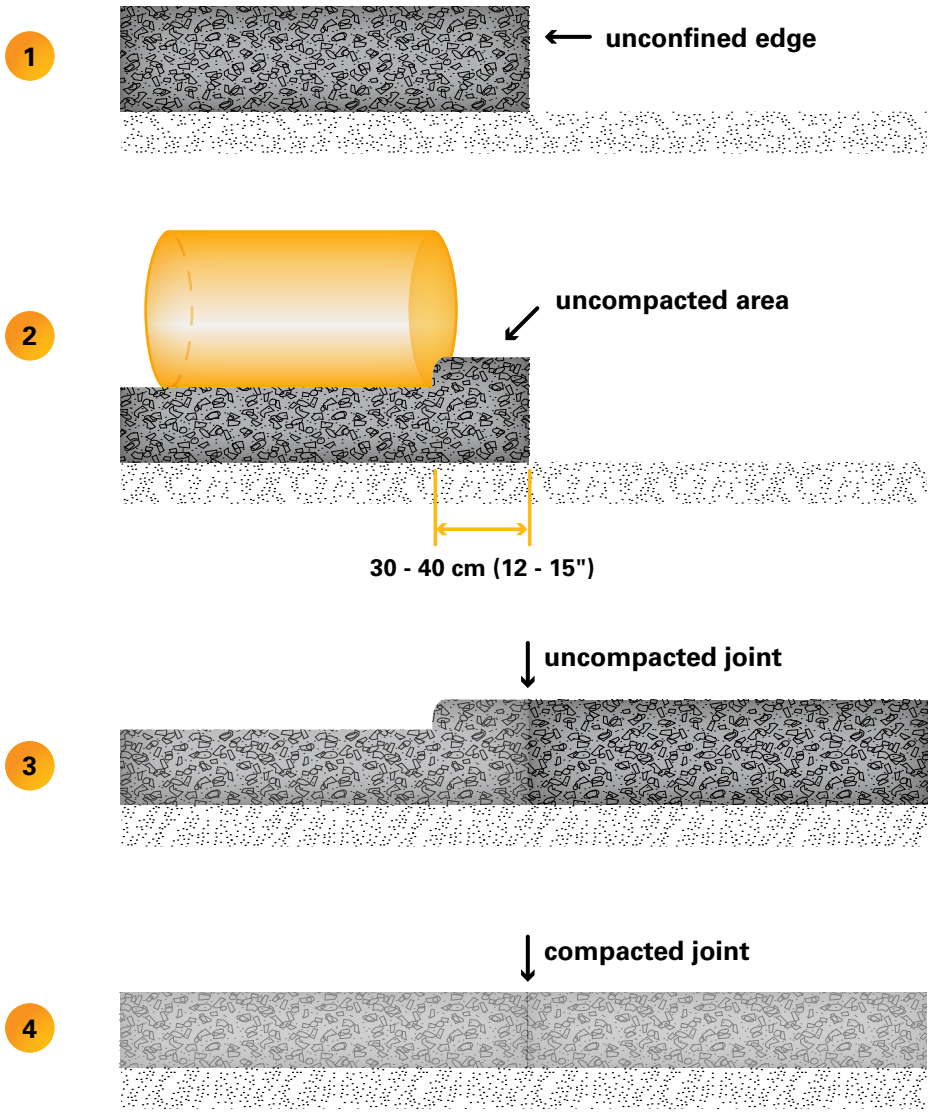
- Extend the pavement 30 to 45 cm (12 to 15") beyond the design width and cut back to fully compacted material after compaction.
- Use a temporary edge restraint.

In general, roller-compacted concrete should be compacted within one hour of mixing. However, at longitudinal joints, it is acceptable to delay compaction up to an additional hour when adjacent layers form a longitudinal joint.



Compacting longitudinal joints requires a special pattern.

CONCRETE ROLLER-COMPACTION SEQUENCE



When possible, compaction of unconfined edges is delayed until the adjacent lane is paved.

The compaction process can begin as soon as the first layer of roller-compacted concrete is started. The compactor(s), however, must stay away from the unconfined edge that will be part of the longitudinal joint.



To avoid deforming the edge, stay away from the edge 30 cm to 40 cm (12" to 15").

During the joint matching pass, the paving crew uses the uncompacted edge of the previous layer as the height reference.

As the matching layer is being paved, start the compaction by straddling the longitudinal joint with half the drum on each side. Depending on the type of material, the depth of the layer, the density developed by the screed and the specified final density, it may be necessary to start with a relatively light compactor and progress to heavier vibratory or pneumatic compactors.



Straddle the longitudinal joint when making the first pass forward.

[PAVING AGGREGATE BASE]

Using a paver to place aggregate base or plant-manufactured, stabilized aggregate base is preferred on some projects because of the paver's ability to provide precise depth and width control. Pavers equipped with combination vibratory and tamping screeds also develop relatively high initial density compared to conventional grading procedures. The depth of the aggregate layer can be up to 30 cm (12") with minimal or no modifications to the screed lift cylinders and the auger lift cylinders. Depending on the density specification and the availability of high-production compaction equipment, laying down multiple, thinner layers may produce better results.

The width of the aggregate layer can be set to the limits of any screed. Often the paving width is determined by the desired production and the paving speed that best matches the target for initial layer density.



2.5 m (8') combination screed laying down aggregate base 3.5 m (11.5') wide and 30 cm (12") deep.

For example, assume a project where the production rate is 500 tonnes (550 tons) per hour. At a layer depth of 25 cm (10"), the 2.5 m (8') combination screed that is planned to be used normally develops around 85 percent density at a working speed of 4 m (13') per minute. The Paving Production Calculator can be utilized to determine the paving width that matches the hourly production.

PAVER SPEED CALCULATOR

Trucking
Paver Speed
Compaction
Windrow
Yield
Slope
Thickness
Job Summary
Legal

Exit

General Inputs	*Imperial Units*	*Metric Units*
Paving Thickness:	[**9.84**] *in*	[**250.0**] *mm*
Paving Width:	[**16.40**] *feet*	[**5**] *meter*
Material Density Uncompacted:	[**120**] *lbs/ft³*	[**1922**] *kg/m³*
Paver Speed @ Given Production Rate		
Production Rate of Hot Plant:	[**551**] *tons/hr*	[**500**] *tonnes/hr*
Calculated Paving Speed - 100% Efficiency:	[**11.4**] *ft/min*	[**3.48**] *m/min*
Calculated Paving Speed - 95% Efficiency:	[**12.0**] *ft/min*	[**3.65**] *m/min*
Calculated Paving Speed - 90% Efficiency:	[**12.5**] *ft/min*	[**3.83**] *m/min*
Calculated Paving Speed - 85% Efficiency:	[**13.1**] *ft/min*	[**4.00**] *m/min*
Calculated Paving Speed - 80% Efficiency:	[**13.7**] *ft/min*	[**4.18**] *m/min*
Calculated Paving Speed - 75% Efficiency:	[**14.2**] *ft/min*	[**4.35**] *m/min*
Effective Paving Speed:	[**11.4**] *ft/min*	[**3.48**] *m/min*

Select the Paver Speed screen. Enter the paving thickness, 250 mm (10"). Enter the material density, 1922 kg/cm³ (120 lb/ft³) in this example. Enter the production rate, 500 tonnes (560 tons) per hour. Then, experiment with the paving width. For this project, assume an efficiency rate of 85 percent based on trucks transferring directly to the paver hopper. A paving width of 5 m (16.4') will be exactly right to produce a working speed of 4 meters (13') per minute.

Or, make calculations to determine the maximum productivity that is possible using a paver to lay down the base aggregate. It may be beneficial to compare the use of a paver against the use of a motor grader. To determine the maximum productivity, the paver's maximum paving width and maximum paving depth must be known. Also, the working speed that will develop the desired initial density must be known.

Again, select the Paver Speed screen. The paver to be utilized has a maximum paving depth of 30 cm (12") and a maximum paving width of 7 m (23'). Enter those values. A working speed of 4 m (13') per minute normally produces acceptable initial density. Assume an 85 percent efficiency rate. Experiment with hourly production until 4 m (13') per minute appears as the calculated paving speed at 85 percent efficiency. In this example, the maximum hourly production rate is 850 tonnes (952 tons). To think in terms of volume rather than weight, that's 440 cubic meters (578 cubic yards) per hour.

PAVER SPEED CALCULATOR

Trucking
Paver Speed
Compaction
Windrow
Yield
Slope
Thickness
Job Summary
Legal

Exit

General Inputs	*Imperial Units*	*Metric Units*
Paving Thickness:	[11.81] *in*	[300.0] *mm*
Paving Width:	[22.97] *feet*	[7] *meter*
Material Density Uncompacted:	[120] *lbs/ft³*	[1922] *kg/m³*
Paver Speed @ Given Production Rate		
Production Rate of Hot Plant:	[937] *tons/hr*	[850] *tonnes/hr*
Calculated Paving Speed - 100% Efficiency:	[11.5] *ft/min*	[3.51] *m/min*
Calculated Paving Speed - 95% Efficiency:	[12.1] *ft/min*	[3.69] *m/min*
Calculated Paving Speed - 90% Efficiency:	[12.7] *ft/min*	[3.86] *m/min*
Calculated Paving Speed - 85% Efficiency:	[13.2] *ft/min*	[4.04] *m/min*
Calculated Paving Speed - 80% Efficiency:	[13.8] *ft/min*	[4.21] *m/min*
Calculated Paving Speed - 75% Efficiency:	[14.4] *ft/min*	[4.39] *m/min*
Effective Paving Speed:	[11.5] *ft/min*	[3.51] *m/min*

Set up the paver with grade and slope control the same way as when placing bituminous material. A single grade sensor located adjacent to the auger chamber will produce the best yield. An averaging ski will produce the best longitudinal profile. Slope control may be used to correct the transverse profile.



Conventional grade and slope control provides precise depth and transverse profile.

Following are five items to remember when paving thick layers of aggregate base.

1. Adjust tow-point to the depth of the layer. When laying down thick aggregate bases, the tow-point will be near the top of its range.

2. Adjust the thickness screws on both sides to give the main screed the correct angle of attack. Turn each screw, so the indicator is in the "plus" range whenever the paving thickness is 15 cm (6") or more.

3. If the paver has two positions for the connection of the auger lift-cylinder rod, select the lower position. The lower position gives the auger additional clearance when the rod is fully retracted.

4. Tow-point modifications may be required to permit laying down thick base layers. Consult the equipment supplier for assistance.

5. Increase the frequency of inspection on wear items. Laying down aggregate base causes more wear on components like conveyors, augers and screed plates. Unlike bituminous mixtures that have lubricating properties from the asphalt cement, water is the only binder for aggregate bases. Therefore, aggregate mixtures are much more abrasive. Expect to see higher wear than normal. To help minimize wear on screed plates, always activate the screed counterbalance system when laying down aggregate mixtures. Also, maintain the correct angle of attack to avoid excessive wear on the trailing edge of the screed plates.











[SUMMARY]

Each paving application has its own requirements. Those requirements must be discussed and understood by the crewmembers prior to paving. What is most important on the application: yield, elevation, smoothness, slope or appearance? In general, the paver can be setup to meet any of these requirements.

If special attachments or tools such as averaging skis are required, be sure to make timely arrangements for their delivery to the project. Some projects have written method specifications that dictate use of certain types of equipment, attachments or techniques. Be sure to follow all mandatory guidelines.

Set up grade and slope controls to satisfy project requirements. Crewmembers should be able make decisions regarding the position of grade sensors to control screed reaction. Stress the importance of mounting grade sensor hardware in a manner that permits the crew to position the sensors where they need to be. Crewmembers should be able to decide whether to use automatic slope control or to make slope corrections manually.

Consider all the project variables when selecting averaging skis. Mechanical averaging skis provide maximum reduction in roughness. Non-contact averaging skis provide the most versatility and convenience.

For special applications, such as roller compacted concrete or deep-lift aggregate base installation, be sure to make the necessary machine modifications.

Finally, lay out each project to maximize efficiency and communicate production goals, as well as quality control concerns to each crewmember.


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Unit 8
TROUBLESHOOTING GUIDE

Troubleshooting can transform a disaster into a routine day at the worksite. Recognizing mat defects, determining their causes, and preventing or correcting them require a team effort involving supervisory personnel, paver and screed operators and quality control. When problems occur, knowing what to look for and how to adjust will help maintain quality, productivity and overall worksite efficiency.



Recognizing common defects in bituminous layers, how to determine the cause of those defects, and how to prevent or correct them is critical to every paving contractor.

Often times, mat defects are the result of crews not executing the fundamentals of paving correctly. Small missteps often result in much larger flaws. When crews perform the fundamentals correctly, many potential issues can be avoided.

This unit covers visible defects as well as other defects such as roughness and the conditions that contribute to the problem.

Paving efficiency dramatically affects the quality of the bituminous material. Continuous paving at a consistent speed makes it easy for the crew to keep all the factors that affect the screed in balance.

Proper use of automatic grade and slope control contributes to quality—not just smoothness or yield, but the appearance of the material as well. If the crew understands the most important project requirements, they should be able to set up grade and slope to meet those requirements. However, they must realize that "automatic" does not mean grade and slope are set up the same way on every project.

Developing knowledge of the causes and cures of paving issues is a continual process. Every project presents an opportunity to use past experiences and gain new knowledge. Equipment changes over time and specifications change, too. What was learned a few years ago, may not be helpful today.

[SURFACE TEXTURE]

The surface texture of the bituminous layer should be uniform from edge to edge across the full width of the layer. The type of material being placed does have an effect on surface texture. Dense-graded mixtures will appear tight and somewhat shiny. Open-graded mixtures will have a more open texture and will look darker. However, no matter what type of mixture is being paved, the texture should be the same to provide the best quality. If texture stripes are visible, use the following checklist of the most common potential causes to identify and correct the problem:

- Cold screed plate
- High spots in the grade
- Compacted spill in front of the paver
- Wavy screed plate
- Screed plates with different lengths
- Strike-off attachments
- Damaged screed
- Incorrect angle of attack
- Augers set too low



A screed that is not properly heated or that has cold sections will create drag marks and texture stripes.

When any portion of the screed plate is significantly colder than the bituminous material passing under it, the hot material will stick to the cold metal. The surface of the bituminous layer will look open or torn in that area. Normally, this will be readily apparent as the paver pulls away from the starting reference.

Stop paving and let the cold screed plates rest on the hot mat for a short period of time before resuming. The hot asphalt will quickly heat them. Cold screed plates also create drag that causes the screed to drop as it pulls away from the starting reference. A depression is often formed and repair may be necessary. Hand repair can take place while waiting for the screed plates to heat.

Another solution for repairing the depression is to return to the starting reference. Heat the screed adequately. Conduct start-up procedures and lower the screed to the starting reference. Start paving again and monitor the mat for drag marks and texture stripes.

User Tip: *Caterpillar recommends leaving the generator activated when paving with an electrically heated screed, especially when paving on urban projects with frequent restarts. The electric screed heating system will automatically energize screed heating elements, if the temperature sensors detect that the screed plates lost too much heat and are below the target temperature.*

On some projects, high spots in the grade may exist. When the screed passes over these high spots, the texture of the asphalt layer will appear open because aggregates are being dragged. This is especially a problem when paving with automatic slope control. As a rule, the thickness of the layer must be at least 1.5 times greater than the largest aggregate size for the mixture to pass under the screed in a uniform manner.

High spots in the grade can be the result of:

- Improper cold planing procedures
- Rutting and shoving
- Improper grade preparation
- Base deformation by haul units

Prior to paving, repair high spots in the grade. Always check grade conditions prior to paving.



Paving over a high spot in the grade will create texture differences in the surface of the asphalt layer.

Texture stripes caused by high spots may appear and disappear. When there is adequate layer thickness and all other factors are equal, the surface texture is normal. However, when the layer gets too thin, as when paving over a high spot, the texture becomes open and torn.



Temporary texture stripes caused by paving over high spot in milled surface.

Some high spots in the grade occur during the paving process when material is spilled and not immediately cleaned up. Spills become major problems when compacted by the haul unit tires or by the paver undercarriage. The compacted spill is no longer loose material, but is a hard, high spot.



Spills in front of the paver can cause high spots in the grade.



Drag mark caused by paving over compacted spill.

Depending on the thickness of the compacted spill, the screed may loose its floating characteristic when paving over the spill. The screed may ride up over the high spot, causing roughness. Drag marks and open texture will be apparent. Open texture surfaces lose heat faster than tight textures. Therefore, the area of open texture will be significantly cooler than the surrounding material. Fractured aggregates in the area over the compacted spill may be visible, because the layer is too thin to absorb vibratory energy. All these factors contribute to roughness, poor density and unacceptable layer thickness.

Texture stripes in the surface of the bituminous layer can be caused by a wavy screed plate. When screed plates are installed, the trailing edge should be checked for flatness. If high points exist, those areas of the screed plate will exert more pressure and the surface will appear tight and shiny. If there are low spots along the trailing edge of the screed, the screed will exert less pressure and open texture stripes will appear in the surface of those areas. If left wavy, eventually the screed plate will wear itself flat. However, the uneven wear pattern will shorten the service life of the screed plate.



A wavy screed plate exerts uneven pressure and creates texture stripes.

User Tip: *Cat dealers can provide clear instructions for replacing screed plates. Ask for "Screed Checks and Adjustments."*

On some screeds, the length of the main screed plate and the length of the extension screed plates are different. For example, the main screed plate may be 45 cm (18") long (front to back) and the extension screed plates may be 30 cm (12") long (front to back). Because the screed plates have different lengths, they exert different pressures on the surface of the layer. The texture differences are normally relatively minor. Adjust the screed extensions' angle of attack to create a more uniform texture.



Screed plates of different lengths can cause minor texture differences.

The screed angle of attack has a major influence on the texture of the bituminous layer surface. If the angle of attack is too high, the surface will have a tight and shiny appearance. If the angle of attack is too low or if the nose of the screed is angled down, the surface will appear open and torn.



Multiple texture stripes caused by multiple factors.

The image above shows examples of an incorrect angle of attack, a damaged screed plate, and open texture behind a strike-off plate.

1. On this project, there is a requirement for a 60 cm (24") wide, 30 degree taper at the edge of the paved surface. The crew forms the taper using an angled strike-off attachment. There is no screed plate to create texture. There is only a strike-off edge. Therefore, the texture will be very open. This is to be expected, and the crew should not take any action.

2. There is a 45 cm (18") wide extension bolted to the end of the left screed extension. The bolt-on extension is running flat. In other words, there is not enough angle of attack. The texture of the surface is much more open, than the texture of the surface toward the center of the layer. The crew should adjust the trailing edge adjusters on the bolt-on extension until the angle of attack is correct and the texture matches the remainder of the layer.

3. The strike-off in front of the bolt-on extension was damaged when it struck a drain inlet. The bent strike-off is causing the darker texture stripe. The crew must repair or replace the bolt-on extension to get rid of that texture stripe.

4. There is also open texture behind the left portion of the left screed extension. This texture stripe may be caused by a wavy screed plate. More likely, it is caused by the screed running with a flat or nose-down angle of attack. Notice the excessive head of material in front of the left extension. Too much weight in front of the extension causes the extension to climb. The grade averaging ski on the left side of the paver will detect this screed movement. The grade control system will constantly be sending signals to the tow-point to move down to compensate for the lifting force created by the head of material. Consequently, the angle of attack is always nosed down. The solution to the problem is lowering the head of material by adding additional auger segments and more mainframe extensions.

This is an example of what can happen when fundamentals are not closely followed. Never pave with a continuously excessive head of material.

When the augers are too low, a distinct texture pattern appears. The surface of the bituminous layer will be tight in the center. There will two areas of open texture on either side of center. Those texture stripes, commonly called "auger shadows," will be the same width as the augers. The edges of the layer will have a tighter texture, similar to the stripe in the center.

To eliminate the auger shadows, increase the auger height until the texture of the surface becomes uniform. Auger shadows are more likely to occur when the mix has large aggregates, 19 mm (0.75") or larger. The rule of thumb is to maintain auger height about 50 mm (2") above the surface of the layer. When paving with large aggregate mixes, the height may need to be increased to 75 mm (3") above the surface of the layer.



Running with the augers too low can cause two areas of open texture directly behind the auger shafts.

[SEGREGATION]

The bituminous material being placed by the paver is supposed to be a uniform blend of aggregates, fine material, asphalt cement, additives, recycled material and so forth according to the mixture formula. For a variety of reasons, the larger aggregates in the mixture sometimes separate from the smaller aggregates and show up as stripes or patches in the bituminous layer. These areas of segregated larger aggregates are susceptible to moisture penetration and usually develop lower than acceptable density. Eventually, areas with segregation may become cracked and end up as potholes.

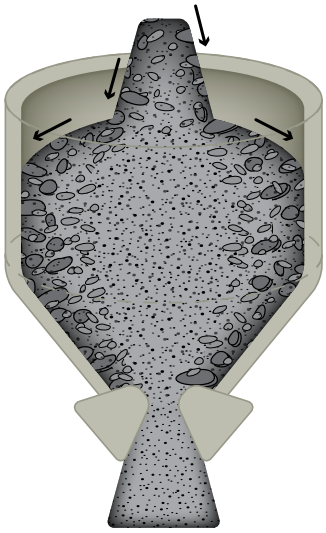
Although material segregation can occur in almost any type of bituminous mixture, mixtures that are the most prone to material segregation are those with aggregates 19 mm (0.75") or larger. Larger aggregates are heavier than their smaller counterparts. Anytime a mixture is transferred, dumped or moved, the larger aggregates can roll away from the smaller aggregates and accumulate as segregated pockets.

Segregation can occur at various places:

- In the storage silo
- In the haul unit
- In the transfer device
- In the paver hopper or hopper insert
- In the auger chamber

Segregation in storage silos can occur as a natural by-product of the silo loading process or can occur if the loading or discharging components are defective.

During silo filling, the contents can form some sort of cone or peak directly under the filling point. Some segregation takes place during the filling as more of the larger aggregates roll further down the slope of the cone toward the walls of the silo. When emptying through a single central outlet, the cone inverts and a hole develops down the center above the outlet. The initial flow of material will show a shift toward fine, and when the hole reaches the diameter of the silo, it will shift back, becoming coarser. When further material is put into the partially emptied silo it will segregate to form a coarse layer (immediately above the level of the partial load) as the cone builds up to the full diameter of the silo. This will show up as a coarse accumulation the next time material is taken from this level. Repeated cycling can build up layers of segregated material.



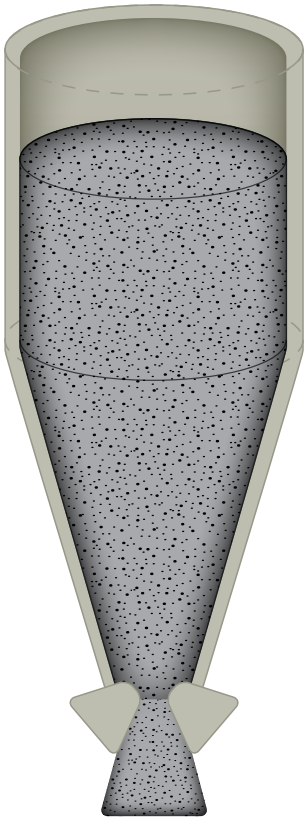
Larger aggregates can roll to the outside of a storage silo during loading.

Material that has been segregated in such a manner in the storage silo will show up as random patches of large segregated aggregates in the bituminous layer. The segregated mix can be detected as it is discharged from the haul unit into the paver or into the transfer device.

Anti-segregation devices (plates, deflectors), as well as rapid-opening clam gates to load batchers and mass-flow, steep-angle discharge systems, help minimize this problem.

Rapid-opening clam gates minimize segregation during the loading of the storage silo. Rather than loading the silo from a conveyor, the mix is transferred to a batch storage bin at the top of the silo. When the batch bin is full, twin clam gates open rapidly and the mix falls as a mass into the silo. The mix flattens on impact and does not flow to the sides of the silo.

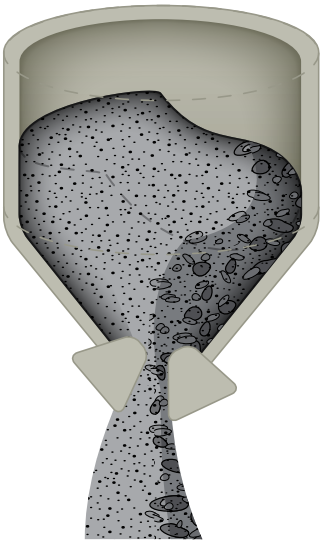
Mass-flow silos have discharge cones with steep angles. The cone angle is usually between 66 and 72 degrees. The mass-flow pattern allows material to move down the silo as a column with no flow channels. There are fewer areas with stagnant, non-moving mix, so the effects of segregation within the storage silo are eliminated or minimized.



Silos with steep discharge angles minimize segregation.

Sometimes, There may be large aggregates in one side of the layer and smaller aggregates in the other side of the layer. This mat defect is usually not created by the paver. Rather, this type of segregation comes from the storage silo. If a clam gate is not opening properly, mix can be directed to one side of the silo. The off-center drop makes larger aggregates roll to the opposite side of the silo. Smaller aggregates and fines end up on the loading side of the silo.

Material that has been segregated in the storage silo will show up either as random patches or as half-and-half segregation. Another type of segregation will show up in a repetitive pattern.



Off-center loading can create segregation on one side of the silo.

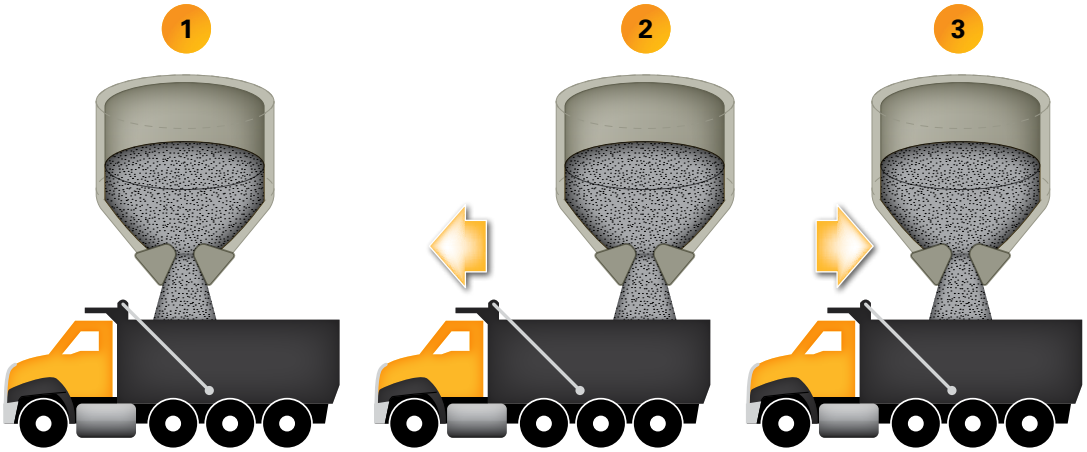
User Tip: *To confirm that the half-and-half segregation is coming from the storage silo, load several haul units from the opposite direction under the silo. If the segregation in the layer changes from side to side when the mix from these trucks is sent through the paver, it can be determined that the segregation is originating in the storage silo.*

End-of-load segregation is usually easy to identify because it appears in the bituminous layer at regular intervals. For example, if the average load in a haul unit will supply the paver for a distance of 30 meters (100'), patches of segregated material will appear every 30 meters (100'). The segregation patches may form a V-shape or may be circular. Their appearance indicates that material is probably segregated in the bodies of the haul units.



End-of-load segregation shows up at regular intervals as patches or V-shapes in the asphalt layer.

How the haul units are loaded at the asphalt plant has a significant impact on the occurence of segregation in the bituminous layer.



Multiple drops help reduce segregation in haul units.

One way to reduce end-of-load segregation is to make sure that trucks are loaded by a three-drop method at the plant. For the first drop, the truck should be positioned so mix fills the front of the truck bed. The front wall of the bed helps confine the mix, so a conical pile is not formed. For the second drop, pull the truck ahead, so the mix falls into the rear of the bed. Now, the tailgate is the vertical barrier that prevents the conical pile. Finally, move the truck back slightly so the third drop is in the center of the bed. The first and second drops help confine the third drop, and there will be much less roll-down of large aggregates.

The three-drop procedure is especially important for mix designs that have a large amount of aggregates 19 mm (0.75") and larger. Fine mixes are less prone to segregation, but the three-drop method is still recommended.

If segregation is not occurring during the truck loading process, continue to troubleshoot the cause of the pattern segregation.

When transferring mix directly into the paver hopper from the haul unit, accumulations of larger aggregates at the sides of the hopper may occur. This phenomenon is an unavoidable consequence of discharging mix from the haul unit into the hopper, especially if there are large aggregates in the mix. Some of the larger aggregates will roll to the sides of the hopper. The key to reducing the amount of segregation in the layer is to stop cycling the hopper wings or, at a minimum, cycle the hopper wings when there is still material in the center of the hopper. Never cycle the hopper wings when the conveyors are low on mix or empty.



Look for segregation in the paver hopper.

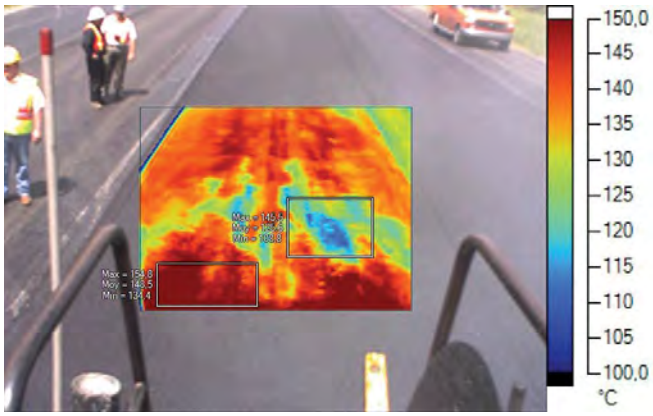
Cycling of the hoppers should only occur while the conveyors are are full of material. Cycling the hopper wings over empty or partially empty conveyors will increase the amount of end-of-load segregation. The problems created by cycling the hopper wings are widely recognized. In fact, some public works agencies have written specifications that prohibit cycling hopper wings on certain types of projects.



Severe segregation in the hopper caused by cycling the hopper wings when the level was too low.

Even if there is no material segregation in the hopper and no visible segregation in the bituminous layer, there can be a large temperature differential caused by cycling the hopper wings. When there is a temperature differential that is 15° C (30° F) or greater, there is a high likelihood of experiencing a significant density variation and also a detectable bump. Some public works agencies have written specifications that require documentation of layer temperature as part of the quality control process. Infrared cameras and on-board heat sensing systems will be more common in the future and can be an important troubleshooting tool.

The use of material transfer devices helps to eliminate or minimize end-of load segregation. However, material transfer devices do not guarantee that segregation can be avoided in the bituminous layer. Several factors need to be monitored.



Large temperature differential caused by cycling hopper wings when the conveyors were almost empty.

Hopper inserts are often used in conjunction with material transfer devices. Depending on the type of mix, the level of mix in the hopper insert and the height from which mix is transferred, a cone forms in the center of the hopper insert and large aggregates roll to the sides of the insert—similar to what can happen when loading a storage silo. When segregation in the hopper insert is observed, be careful not to run the material level low in the insert between trucks.



Segregation can occur in the hopper insert.

Segregated large aggregates that have accumulated around the outer diameter of the hopper insert will pass through the paver as a mass when the level in the insert gets too low. Try to keep the insert at least half full to minimize segregation caused by large aggregates that are the last fraction of the load to transfer out of the insert. Random patch segregation can occur when material in the transfer vehicle empties.



Random patch segregation caused by running the mix level low in hopper insert.

User Tip: *Use the Paving Production Calculator to determine the amount of time that is available to pave during truck exchanges. Using half the volume of the transfer vehicle and half the volume of the hopper insert as the amount that can be consumed while exchanging trucks, determine how far that volume will pave (yield). For example, the amount of mix available may yield 30 meters (100') of paving. If the paving speed is 6 meters per minute (20 feet per minute), there are five minutes of paving time before running the hopper insert and / or transfer vehicle so low that segregated material may begin to pass through the paver.*

It is better to stop the paver and wait for mix rather than continue paving and risk introducing segregated material into the layer. It is also better to stop paving at the end of the shift or the end of a paving pass before consuming all the mix in the transfer vehicle. Keep an empty haul unit at the project. Load the last few tons out of the transfer vehicle into the empty truck. Remove any portion of the bituminous layer that has segregated material from the last portion of the mix adjacent to the hopper insert.

When windrow paving, there may be, segregated aggregates at the end of the windrow. The last material out of the body of belly-dump trucks can be a mass of large aggregates. The dump person should try to feather out the remaining material from the truck. Then, overlap the end of the windrow so the segregated mix is covered with good quality mix. Some blending will occur as the windrow elevator picks up the material.



The end of a windrow often consists of segregated aggregates.

Improper adjustments to the feeder system may cause other segregation issues.

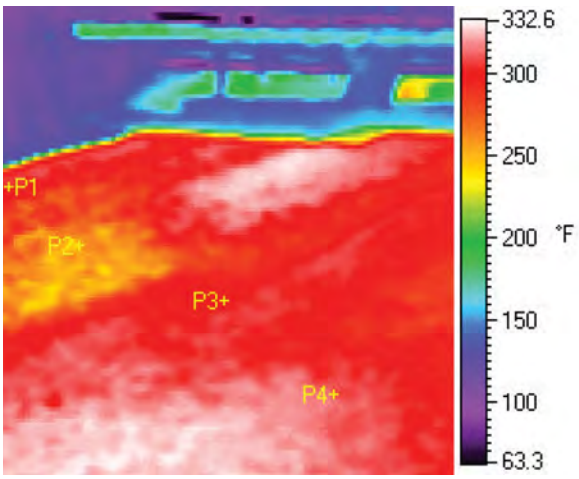
Auger speed has an impact on the appearance of the bituminous layer. The proper auger speed is between 20 and 40 revolutions per minute. When the auger speed is too high, the augers can actually cause the aggregates in the mix to separate. The larger aggregates will roll and stop in the stagnant areas under auger bearing hangers or at the end of the tractor frame. These will be continuous stripes if the augers are turning too fast all the time. If the auger speed varies between an acceptable range and a speed that is too fast, the stripes will come and go.

To slow down the augers, increase the conveyor speed or raise the flow gates. Remember, each conveyor and each auger has individual adjustments. It is possible for one side of the feeder system to function in the acceptable range and the other side to function improperly. There may be a segregation stripe inline with the the right auger bearing support and not on the left side of the mat. This condition indicates that the right side feeder system needs to be adjusted to create acceptable auger speed.



Continuous segregation stripes often relate to high auger speed.





Random patch segregation can be seen and also detected in thermal images.

When segregated aggregates appear occasionally in the layer, that condition is referred to as random patch segregation. Random patch segregation is usually associated with the feeder system operating erratically. In other words, both augers or either auger may rotate at the recommended speed for a period of time and then slow down or stop. When an auger slows or stops, the largest aggregates tend to roll toward stagnant areas. A pocket of segregated aggregates is formed and those aggregates show up as a patch of segregation in the layer. The segregation stops when the augers are rotating at a normal, continuous speed. Hence, the appearance of the segregation is random.

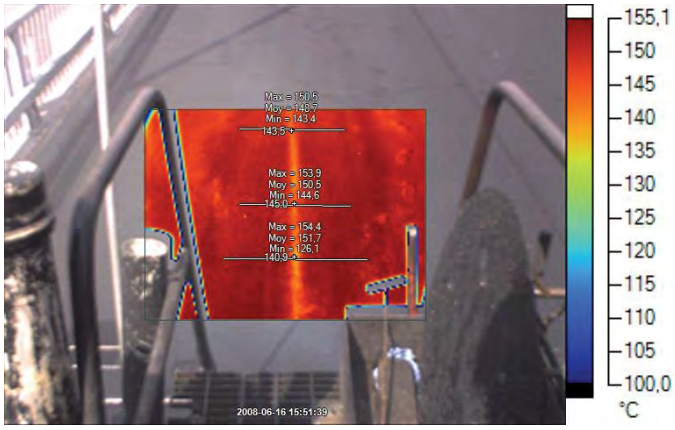
If the augers operate at variable speeds, reposition the feeder sensor. Erratic operation of the feeder system results when the sound pulses emitted by the feeder sensor do not reflect back to the sensor.

Use a tape measure or folding ruler extended 45 cm (18"). Hold the ruler on the end of the sonic feeder sensor. Be sure that the target is about 45 cm (18") away from the sensor and that the sensor is perpendicular to the target, so the sound pulses will reflect back. Position the sensor until the material-height dial and/or conveyor speed dial have the ability to produce consistent feeder system operation on both sides of the paver.



Centerline stripes can be related to feeder system operation.

Segregation stripes in the center of the bituminous layer can result from a variety of causes. The stripe sometimes occurs when the area under the auger chain case restricts material flow. The larger aggregates tend to accumulate in this area and will show up as a stripe, sometimes barley detectable by sight.



A thermal image confirms the open-textured, segregation stripe in the center.

Any area of segregation in the bituminous layer will have a more open texture compared to other surface areas of the layer. The open-textured surface loses heat more rapidly than those areas where the surface is tighter. Using an infrared camera, it is possible to confirm the exact location of the segregation stripe and the severity of the temperature differential.



The head of material under and in front of the center chain case can be stagnant.

The auger drive chain case extends into the auger chamber. The chain case impedes the flow of material to the center of the main screed. To improve the flow of material under the chain case, there are various types of paddles or segments installed on the inner ends of the left and right augers. On some pavers, these segments are installed so both push material under the chain case to eliminate the possibility of voids in the center of the auger chamber. When both segments are pushing material under the chain case, the head of material in front of the chain case may be stagnant. In other words, the material is trapped there, and heat loss along with a possible accumulation of larger aggregates may occur. In this situation, change one of the center auger segments to create a flow of material that has one segment pushing material under the center chain case while the other segment is pulling material away from the chain case. The head of material in front of the chain case will become active, and the segregation stripe should disappear while the temperature uniformity improves.

Inspect these reversing augers or kicker paddles frequently. They experience more wear and damage than the other auger segments. Excessive wear can also cause variations in material flow and stripes.

Don't let material build up on components in the center of the auger chamber. The deflector plates in several locations are designed to help the flow of material and prevent segregation. At the end of the shift, inspect these plates and clean them to prevent the accumulation of cold bituminous material.

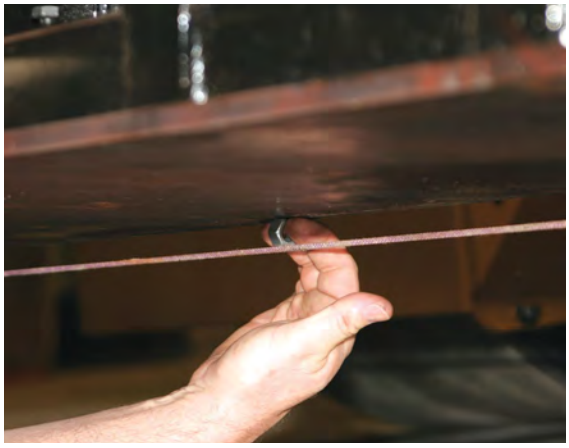
Because the area in the center of the auger chamber can have restricted flow of material, how the mix passes under the center of the screed can be affected. In particular when paving with stiff mixes that have large aggregates, a centerline stripe may appear due to low lead crown.



A centerline stripe can be caused by insufficient lead crown.

Most screeds are adjusted to be flat across the leading edge of the main screed plate as well as the trailing edge of the main screed plate. To help eliminate or reduce centerline stripes, Caterpillar recommends installing 3 mm (0.125") of lead crown into the main screed plate.

To install lead crown, first raise the main screed plate. With the screed raised and securely blocked, place a string across the leading and trailing edges of the screed plate. Use the power crown switch or manual crown adjuster to create a positive crown. Create a 3 mm (0.125") gap between the stringline and the leading edge of the screed plate in the exact center of the screed plate. Then use the trailing edge adjusters at the rear the screed plate to flatten the trailing edge. The leading edge will retain the 3 mm (0.125") lead crown. Confirm that the lead crown has the desired effect of reducing or eliminating the centerline stripe. If the center of the bituminous layer is tight and shiny compared to the rest of the layer, reduce or completely remove the lead crown.



Installing 3 mm (0.125") lead crown in the main screed plate helps reduce centerline striping.